EOSC 213: Computational Methods in Geological Engineering

Lecture 18: Local and global truncation errors

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Motivation

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But an important question is:

How accurate are these methods?

- Numerical methods are approximations
- Approximations introduce error
- We must understand and quantify this error

Two types of error

Two key concepts

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- **Local truncation error (LTE)**
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Big picture

- LTE: error in a single step
- GTE: accumulated error over many steps

Setup: ODE

$$\frac{dy}{dt} = f(t, y), \quad y(0) = y_0$$

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$$y(t_{n+1}) \neq y_n + hf(t_n, y_n)$$

This difference is the **error**.

Where does the error come from?

Taylor expansion:

$$y(t_{n+1}) = y(t_n) + hy'(t_n) + \frac{h^2}{2}y''(t_n) + \cdots$$

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Key idea

Error comes from ignoring higher-order terms.

Local truncation error (LTE)

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Hence, the leading term for Forward Euler LTE is of the order h^2.

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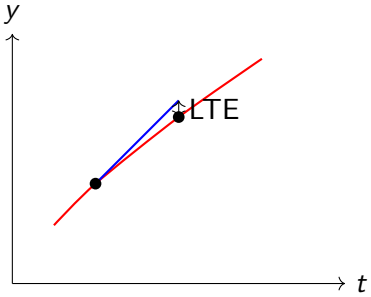
Result

Forward Euler: $\text{LTE} = \mathcal{O}(h^2)$

Visualizing LTE



One-step error starting from the exact solution

Example: Computing local truncation error

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We now compute the local truncation error for Forward Euler.

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Expand:

$$y(t_{n+1}) = \frac{t_n^2 + 2t_n h + h^2}{2}$$

$$= \frac{t_n^2}{2} + t_n h + \frac{h^2}{2}$$

Step 2: Euler approximation

Forward Euler:

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Here:

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So,

$$y_{n+1} = \frac{t_n^2}{2} + ht_n$$

Step 3: Compute LTE

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Substitute:

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Conclusion

Local truncation error is proportional to h^2

What does this result mean?

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This assumes we start from the **exact solution**.

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This assumes we start from the **exact solution**.

But in practice, we never do...

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In reality:

- We do NOT start each step with exact $y(t_n)$

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Key idea

Errors **accumulate over steps**

How many steps do we take?

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Important

As h decreases, the number of steps increases.

Accumulation of local errors

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Conclusion

Global truncation error is $\mathcal{O}(h)$

Important clarification

Why not exactly $\frac{1}{h} \times h^2$?

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Takeaway

Order analysis focuses on how error scales with h, not exact constants.

Example: Global truncation error

Use the same ODE:

$$\frac{dy}{dt} = t, \quad y(0) = 0$$

Exact solution:

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Forward Euler update:

$$y_{n+1} = y_n + ht_n$$

Compute numerical solution

Start with:

$$t_0 = 0, \quad y_0 = 0$$

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Step 3:

$$y_3 = y_2 + h(2h) = h^2 + 2h^2 = 3h^2$$

Compute numerical solution

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Pattern emerges.

General expression

We get:

$$y_n = h^2(0 + 1 + 2 + \cdots + (n-1))$$

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So:

$$y_n = \frac{h^2}{2} n(n-1)$$

Compare with exact solution

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Difference:

$$\text{GTE} = y(t_n) - y_n = \frac{h^2}{2}(n^2 - n(n-1))$$

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Difference:

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$$= \frac{h^2}{2}(n)$$

Final result for GTE

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Conclusion

Global truncation error scales like $\mathcal{O}(h)$

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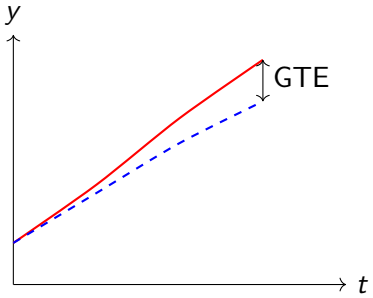
Result

Forward Euler: $\text{GTE} = \mathcal{O}(h)$

Visualizing GTE



Accumulated error over many steps

Connection between LTE and GTE

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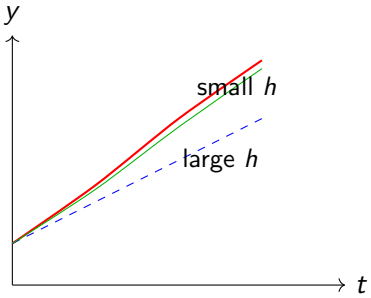
Key insight

Global error is one order lower than local error.

Effect of step size



- Smaller $h \Rightarrow$ more accurate

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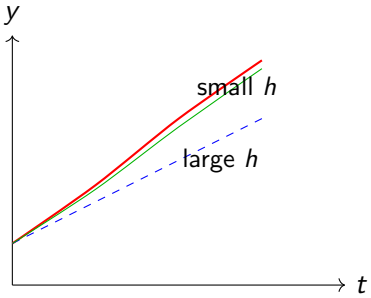

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- Larger $h \Rightarrow$ less accurate

Summary

- Local truncation error (LTE)
  - One-step error
  - $\mathcal{O}(h^2)$ for Forward Euler
- Global truncation error (GTE)
  - Accumulated error over many steps
  - $\mathcal{O}(h)$ for Forward Euler
- Step size controls accuracy

Takeaway

Understanding error is essential for reliable numerical methods.